

MALAK GAMAL EL-DIN

Irvine, CA 92618 $\diamond$ malakgamal7@gmail.com $\diamond$ [linkedin](#) $\diamond$ 949-881-7086 $\diamond$ Citizenship: USA

EDUCATION

University of California, Irvine

Fall 2025 – Present

Ph.D. and concurrent M.S. in Computer Engineering

Research Focus: Edge AI, Neurosymbolic AI, Hardware–Software Co-Design

NSF Graduate Research Fellowship Program (GRFP)

Fall 2026 – 2030

GPA: 4.0

University of Western Ontario, Canada

October 2024

B.E.Sc. in Computer Engineering, Specialization in Electronic Devices for Ubiquitous Computing

GPA: 4.0

Honors: GM Canada Scholarship of Distinction in Engineering, Dean’s Undergraduate Research Award, UWO

In-Course Scholarship, Dean’s List, Western Scholarship of Excellence

PUBLICATIONS

M. Gamal El-Din et al., “Bern2Edge: An End-to-End Neurosymbolic Compiler for Efficient Edge Deployment via Bernstein Polynomial Networks,” *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2026. Accepted for presentation at IEEE/ACM CODES+ISSS, ESWEEK 2026.

I. Albool, **M. Gamal El-Din**, et al., “From Dead Neurons to Deep Approximators: Deep Bernstein Networks as a Provable Alternative to Residual Layers,” *arXiv preprint arXiv:2602.04264*, 2026.

M. Gamal El-Din et al., “Deep Learning-Driven Dual-Axis Tilting Solar Panel System for Enhanced Energy Efficiency and Sustainability,” in *2025 IEEE Electrical Power and Energy Conference (EPEC)*, pp. 110–116, 2025.

EXPERIENCE

Graduate Student Researcher

Fall 2025 – Present

University of California, Irvine

Irvine, CA

- Conducting research in Edge AI, hardware–software co-design, and efficient machine learning for resource-constrained computing platforms.
- Developing hardware-aware neural architectures and compiler/HLS frameworks to reduce inference latency, memory usage, and computational overhead on edge devices and FPGAs.
- Developed Bernstein-based deep neural architectures using learnable polynomial activations to improve trainability and approximation capacity without residual connections.
- Developed an end-to-end neurosymbolic compiler for FPGA edge deployment using knowledge distillation, quantization, HLS code generation, and symbolic rule extraction to reduce latency and memory utilization.

DevOps Software Engineer

September 2022 – May 2023

CGI Inc.

Toronto, ON, Canada

- Built CI/CD pipelines using Python, Jenkins, and Oracle Cloud Infrastructure and maintained 99% uptime across 40+ Linux servers, reducing deployment operation time by 20–30%.
- Re-engineered CSA’s financial regulatory system, improving transaction security and reducing operation costs by 30%; configured TCP, UDP, HTTP, TLS/SSL, and load balancing.

Software Developer

October 2021 – September 2022

University of Western Ontario

London, ON, Canada

- Contributed to an Epic Games–funded medical rehabilitation project, developing VR/AR environments in Unreal Engine (C++) for Microsoft HoloLens and generating 3D models from patient data.
- Developed a ReactJS/JavaScript frontend and Node.js/Java backend for an AR-based hand telerehabilitation platform.

SKILLS

- **Languages:** Python, C, C++, Java, JavaScript, SQL, MATLAB, VHDL, Shell.
- **ML & Hardware:** PyTorch, TensorFlow, HLS, CUDA, Neurosymbolic AI, Knowledge Distillation, Reinforcement Learning.
- **Systems & Tools:** ROS, SLURM, Conda, Kubernetes, Docker, OCI, Jenkins, CI/CD.